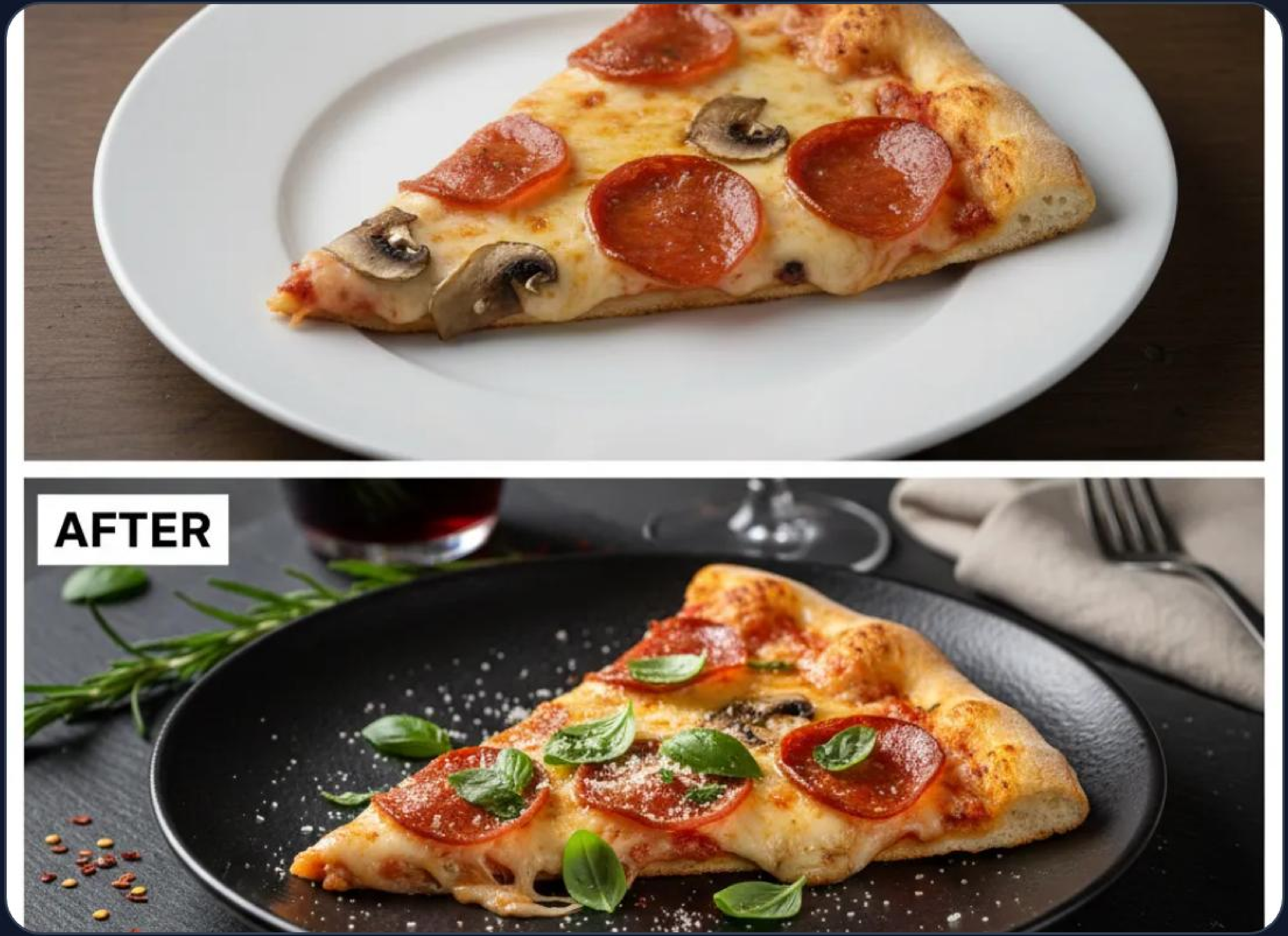
MySQL Data Analytics Portfolio

Pizza Sales SQL Analysis

Exploring Revenue Trends, Category Performance, and Customer Preferences

PROJECT MISSION



Objective

This project leverages MySQL to transform raw sales data into actionable business intelligence. We aim to identify the highest revenue generators and optimize the pizza menu strategy.

- Revenue Trend Discovery
- Category Performance Tracking
- Portfolio Ready SQL Showcase

DATABASE SCHEMA

pizza_types

Stores descriptive metadata: Category (Veggie, Meat, etc.) and Name.

pizzas

Contains Size variations (S, M, L) and their specific price points.

order_details

Transaction records linking orders to specific pizzas and quantities.

MODERN TECHNICAL STACK

Engine: MySQL 8.0

Utilized advanced SQL features to aggregate over 48,000+ records across four relational tables.

- **Relational Joins:** Multi-level table connections.
- **Aggregations:** Complex SUM() and COUNT() metrics.
- **Window Functions:** RANK() and OVER() for segmentation.

RELATIONAL LOGIC

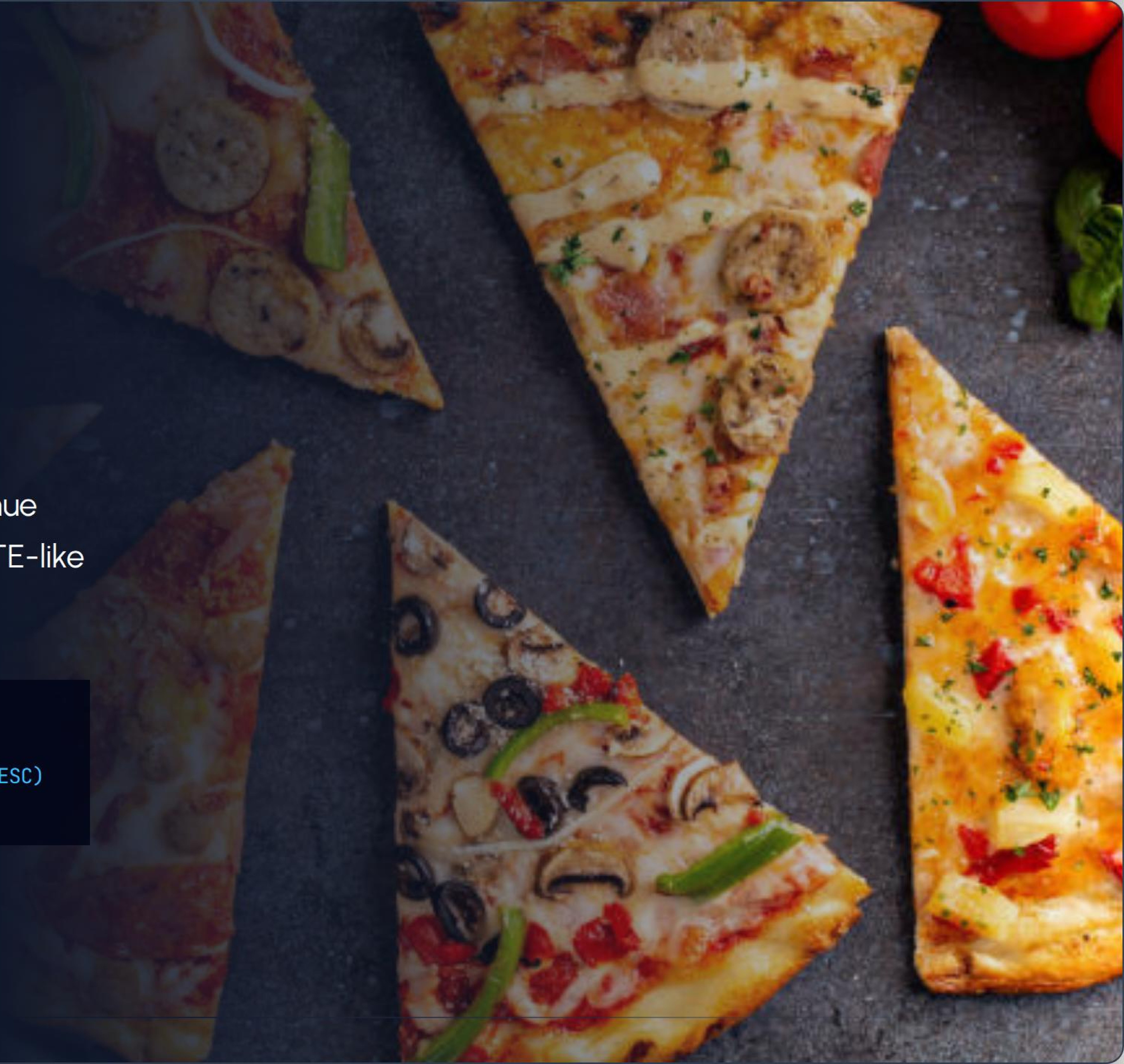
Primary Keys & FKs

Established strong referential integrity between `pizza_id` and `order_id` to ensure data consistency across the sales lifecycle.

Top 3 Pizzas by Category

An advanced SQL challenge to rank pizza revenue performance within specific categories using CTE-like subqueries and window functions.

```
SELECT name, revenue FROM (
  SELECT category, name, revenue,
  RANK() OVER(PARTITION BY category ORDER BY revenue DESC)
AS rn...
```



SQL SYNTAX BREAKDOWN

```
SELECT name, revenue FROM (
    SELECT category, name, revenue,
    RANK() OVER(PARTITION BY category ORDER BY revenue DESC) AS rn
    FROM (
        SELECT pizza_types.category, pizza_types.name,
        SUM((order_details.quantity) * pizzas.price) AS revenue
        FROM pizza_types
        JOIN pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
        JOIN order_details ON order_details.pizza_id = pizzas.pizza_id
        GROUP BY pizza_types.category, pizza_types.name
    ) AS a
) AS b WHERE rn <= 3;
```

Key Implementation: Nested subqueries to first calculate revenue, then apply the RANK() window function per category.

REVENUE DISTRIBUTION

The SQL analysis identified the top performers contributing to 60%+ of total category revenue.

IMPACT HIGHLIGHTS

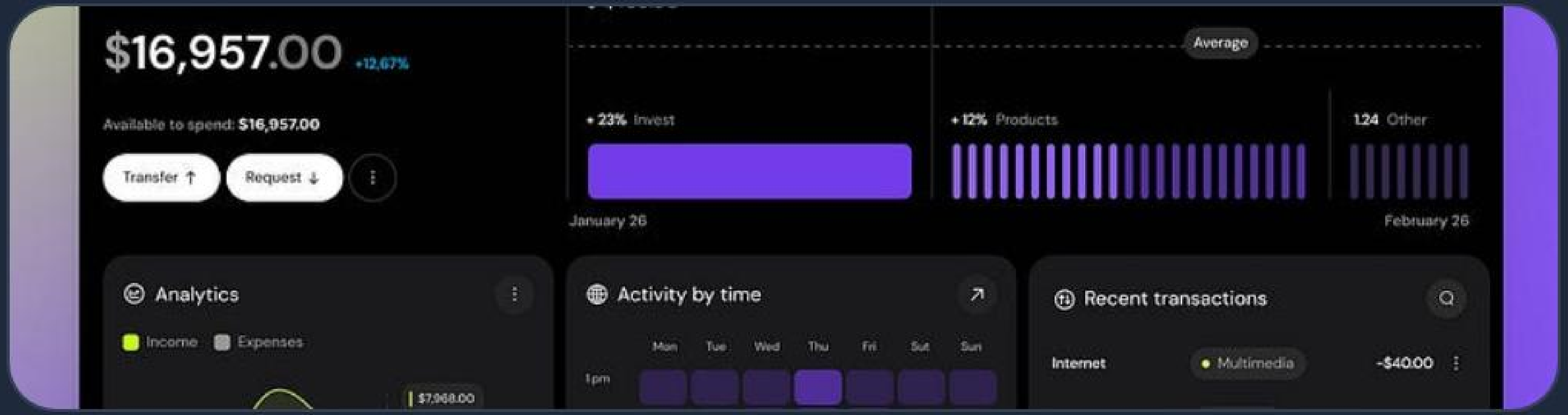
48k+
TOTAL QUANTITIES

4
MAIN CATEGORIES

Top 3
PIZZAS PER TYPE

Analysis revealed that **Chicken** and **Classic** categories dominate total revenue, while **Veggie** pizzas maintain the highest order frequency during lunch hours.

CORE SKILLS SHOWCASED



Business Logic

Extracting insights that drive menu optimization and inventory management.

Quality Code

Writing clean, readable, and documented SQL queries for GitHub repositories.

Ready for Collaboration

This project demonstrates my ability to navigate complex relational databases and extract meaningful business metrics.

View the Full Repository on GitHub

github.com/portfolio/pizza-sales-sql

Data is the raw material of the 21st century. It is the new oil, but only if you have the right tools to refine it.

— SQL Analysis Project Conclusion

IMAGE SOURCES

https://www.aimenuphoto.com/images/guides/ai-prompts-food-photography/hero-image.webp

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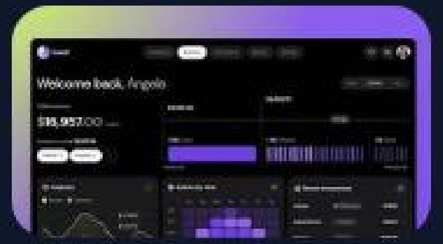

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